

$^{12}\text{C}(^{28}\text{Si},\alpha p\gamma)$ 2007Ks01,2013Bi10,2020Ra14

2007Ks01: two different experiments: one with $E=70$ MeV ^{28}Si beam at BARC-TIFR facility and the second with $E=88$ MeV ^{28}Si beam at NSC facility, both using a target of about $50\text{ }\mu\text{g}/\text{cm}^2$ ^{12}C on a about $10.5\text{ mg}/\text{cm}^2$ gold backing. γ rays were detected with almost identical (INGA) arrays eight Compton-suppressed Clover HPGe detectors. Measured $E\gamma$, $I\gamma$, $\gamma\gamma$ -coin, $\gamma\gamma(\theta)(\text{DCO})$, $\gamma\gamma(\text{lin pol})$, Doppler-shift attenuations. Deduced levels, J , π , $T_{1/2}$, γ -ray multipolarities and mixing ratios. Comparisons with available data and large-scale shell-model calculations.

2013Bi10: $E=110$ MeV ^{28}Si beam was produced from the IUAC-New Delhi accelerator facility. Target was $50\text{ }\mu\text{g}/\text{cm}^2$ thick ^{12}C on $18\text{ mg}/\text{cm}^2$ Au backing. γ rays were detected with the INGA array of thirteen Compton-suppressed Clover detectors. Measured $E\gamma$, $I\gamma$, $\gamma\gamma$ -coin, $\gamma\gamma(\theta)(\text{DCO})$, Doppler-shift attenuations. Deduced levels, J , π , $T_{1/2}$, evidence for an SD band, γ -ray multipolarities. Comparison with large-scale shell-model calculations, two-level mixing calculation, and with structure of SD states in ^{36}Ar .

2020Ra14: $E=110$ MeV ^{28}Si beam was produced from the 15UD Pelletron accelerator at the Inter-University Accelerator Centre (IUAC), New Delhi. Target was $50\text{ }\mu\text{g}/\text{cm}^2$ ^{12}C evaporated onto $18\text{ mg}/\text{cm}^2$ Au backing. γ rays were detected with a multi-detector array (Indian National Gamma Array) consisting of 13 Compton-suppressed clovers. Measured $E\gamma$, $I\gamma$, $\gamma\gamma$ -coin, $\gamma\gamma(\text{DCO})$, $\gamma\gamma(\text{ADO})$, $\gamma\gamma(\text{IPDCO})$. Deduced levels, J , π , γ -ray multipolarities. Comparisons with theoretical calculations.

Other reaction:

1982Le04: $^{28}\text{Si}(^{12}\text{C},X\gamma)^{12}\text{C}$ ($E_{\text{cm}}=23\text{--}35$ MeV) beam produced from the University of Washington FN Tandem Van de Graaff.

A target of $100\text{ }\mu\text{g}/\text{cm}^2$ ^{28}Si on a thick tantalum backing. A Ge(Li) detector. Measured $\sigma(E)$, $E\gamma$.

1985Kr21: $^{27}\text{Al}(^{12}\text{C},X)$ $E=85$ MeV/nucleon. Measured particle-particle angular correlations.

 ^{35}Cl Levels

$E(\text{level})^\dagger$	$J\pi^\ddagger$	$T_{1/2}^\#$	Comments
0.0	$3/2^+$		
1763.0 6	$5/2^+$		
2645.8 6	$7/2^+$	<0.90 ps	
3003.1 8	$5/2^+$		
3163.1 6	$7/2^-$		
3942.0 7	$9/2^+$	<0.35 ps	
4347.8 6	$9/2^-$	0.91 ps $+2I-16$	
5407.4 8	$11/2^-$	<0.76 ps	
5927.1 7	$(11/2)^-$	<0.28 ps	
6087.6 8	$13/2^-$		
6119.5? 13	$(11/2^-)$		
6660.2 8	$(11/2^-)$		
7873.3 8	$13/2^+$		
8319.4 ^a 9	$15/2^-$	$12.5\text{ }\&\text{ fs }7$	Tentative 8324, $(13/2^-)$ level in 2007Ks01 is considered the same level here by the evaluators. $T_{1/2}$: other: <70 fs (2007Ks01). $Q(\text{transition})=0.65$ 2, $\beta_2=0.30$ (2013Bi10).
8487.8 10	$15/2^-$	<0.07 ps	Tentative 8481, $(15/2^-)$ level in 2007Ks01 is considered the same level here by the evaluators.
8558.2? 13	$(13/2^-)$	<0.14 ps	
8787.3? 13	$(15/2^+)$	<0.28 ps	
8845.1 [@] 9	$17/2^+$		
10181.1 ^a 10	$19/2^-$	$42.3\text{ }\&\text{ fs }28$	Tentative 10181, $(19/2^+)$ level in 2007Ks01 is considered the same level here by the evaluators. $T_{1/2}$: other: <0.14 ps (2007Ks01). $Q(\text{transition})=0.82$ 6, $\beta_2=0.37$ (2013Bi10).
10223.2? 14	$(17/2^-)$	<0.28 ps	
10862.2? 14	$19/2^+$	<0.28 ps	
11459.2 [@] 12	$21/2^+$	$43.0\text{ }\&\text{ fs }21$	$Q(\text{transition})=0.56$ 2, $\beta_2=0.27$ (2013Bi10).
12572.2 ^a 12	$23/2^-$	$33\text{ }\&\text{ fs }4$	$Q(\text{transition})=0.71$ 5, $\beta_2=0.32$ (2013Bi10).
16306.4 ^a 16	$(27/2^-)$	$<9.0\text{ }\&\text{ fs}$	$Q(\text{transition})>0.5$, $\beta_2=0.24$ (2013Bi10).

Continued on next page (footnotes at end of table)

$^{12}\text{C}(^{28}\text{Si},\alpha p\gamma)$ **2007Ks01,2013Bi10,2020Ra14 (continued)** ^{35}Cl Levels (continued)[†] From a least-squares fit to γ -ray energies, assuming $\Delta E\gamma=1$ keV in the fitting procedure.[‡] From Adopted Levels, unless otherwise noted. Assignments for tentative levels not adopted in Adopted Levels are proposed by **2007Ks01**.# From DSAM in **2007Ks01**, unless otherwise noted.@ Existence of $17/2^+$ and $21/2^+$ levels provides indirect evidence in favor of SD band interpretation and parity-doublet cluster structure (**2013Bi10**).& From DSAM in **2013Bi10**.^a Band(A): SD band. Proposed configuration= $3p-3h$, $(sd)^{16}(pf)^3$. Negative-parity partner of SD band in ^{36}Ar (**2013Bi10**). $\gamma(^{35}\text{Cl})$ Δ_{IPDCO} corresponds to integrated polarization asymmetry measurement at $E=70$ MeV at TIFR facility (**2007Ks01**). The values are from e-mail reply of Dec 1, 2006 from one of the authors (M. Saha Sarkar) of **2007Ks01** to the previous evaluator B. Singh. Δ_{IPDCO} and DCO under comments are from **2007Ks01**, unless otherwise noted. Positive value of Δ_{IPDCO} indicates electric character and negative value indicates magnetic character. Expected value of DCO is ≈ 1.0 for a stretched transition gated by a transition of same multipolarity, ≈ 2.0 for dipole when gated by a stretched quadrupole (DCO(Q)) and ≈ 0.5 for stretched quadrupole when gated by a stretched dipole (DCO(D)) (**2007Ks01**). Values deviating from above values indicate a mixed transition. DCO values are also available in **2020Ra14**, but expected values are opposite to those in **2007Ks01** for stretched dipole and quadrupole transitions when gated by a transition of different multipolarity, that is DCO(Q) ≈ 0.5 for stretched dipole and DCO(D) ≈ 2.0 for stretched quadrupole (**2020Ra14**).For ADO ratios under comments from **2020Ra14**, expected values are 0.6 for pure dipole transitions and 1.6 for pure quadrupole transitions or $\Delta J=0$ transitions (**2020Ra14**).

E_γ [†]	I_γ [‡]	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. [#]	δ [#]	Comments
160 [@]	1.1 3	3163.1	$7/2^-$	3003.1	$5/2^+$			I_γ : other: 0.6 2 at $E=70$ MeV.
518	9.2 5	3163.1	$7/2^-$	2645.8	$7/2^+$	D		I_γ : other: 10.0 2 at $E=70$ MeV. DCO(Q)=0.8 1. DCO(D)=1.5 2, $R_{\text{ADO}}=1.6$ 4 (2020Ra14).
526		8845.1	$17/2^+$	8319.4	$15/2^-$			
680	32 1	6087.6	$13/2^-$	5407.4	$11/2^-$	M1(+E2)	+0.1 1	I_γ : other: 8.6 2 at $E=70$ MeV. DCO(Q)=1.3 1 at $E=70$ MeV. $\Delta_{\text{IPDCO}}=-0.09$ 2.
712 ^{@&}	<1	6119.5?	$(11/2^-)$	5407.4	$11/2^-$			
779 ^{@&}	<1	3942.0	$9/2^+$	3163.1	$7/2^-$			
883	3.2 2	2645.8	$7/2^+$	1763.0	$5/2^+$			$I_\gamma=4.6$ 2 at $E=70$ MeV. $\Delta_{\text{IPDCO}}=+0.01$ 5.
914 ^{&}	2.4 5	8787.3?	$(15/2^+)$	7873.3	$13/2^+$	D(+Q)	-0.2 2	DCO(Q)=2.2 6 at $E=88$ MeV.
972	33 1	8845.1	$17/2^+$	7873.3	$13/2^+$	E2		I_γ : other: 2.0 1 at $E=70$ MeV. DCO(Q)=1.0 2 at $E=70$ MeV, 0.9 1 at $E=88$ MeV. $\Delta_{\text{IPDCO}}=+0.08$ 4. DCO(D)=1.8 2, DCO(Q)=0.9 1, $R_{\text{ADO}}=1.6$ 2 (2020Ra14).
1060	7.3 7	5407.4	$11/2^-$	4347.8	$9/2^-$	M1+E2	+0.2 1	I_γ : other: 6.9 3 at $E=70$ MeV. DCO(D)=0.9 1. $\Delta_{\text{IPDCO}}=-0.10$ 4.
1113		12572.2	$23/2^-$	11459.2	$21/2^+$			
1185	25 1	4347.8	$9/2^-$	3163.1	$7/2^-$	M1+E2	-0.3 1	I_γ : other: 25 1 at $E=70$ MeV. DCO(Q)=2.0 2 at $E=70$ MeV, 2.8 3 at $E=88$ MeV. $\Delta_{\text{IPDCO}}=-0.08$ 3.

Continued on next page (footnotes at end of table)

$^{12}\text{C}(^{28}\text{Si},\alpha p\gamma)$ **2007Ks01,2013Bi10,2020Ra14** (continued) $\gamma(^{35}\text{Cl})$ (continued)

E_γ [†]	I_γ [‡]	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. [#]	δ [#]	Comments
1213		7873.3	13/2 ⁺	6660.2	(11/2 ⁻)			
1296	0.4 2	3942.0	9/2 ⁺	2645.8	7/2 ⁺			I_γ : other: 0.2 1 at E=70 MeV.
1336	6 1	10181.1	19/2 ⁻	8845.1	17/2 ⁺	D(+Q)	-0.2 2	DCO(Q)=2.0 4 at E=88 MeV.
1378 @&	2 1	10223.2?	(17/2 ⁻)	8845.1	17/2 ⁺			
1400	<1	3163.1	7/2 ⁻	1763.0	5/2 ⁺			
1579	18 2	5927.1	(11/2 ⁻)	4347.8	9/2 ⁻	(M1+E2)	-0.2 1	I_γ : other: 12 1 at E=70 MeV. DCO(D)=0.9 1 at E=70 MeV, 1.0 1 at E=88 MeV. $\Delta\text{IPDCO}=+0.02$ 3.
1693		10181.1	19/2 ⁻	8487.8	15/2 ⁻			
1702	11.4 6	4347.8	9/2 ⁻	2645.8	7/2 ⁺	E1+M2	+0.5 +5-3	I_γ : other: 10.8 5 at E=70 MeV. DCO(Q)=1.0 2 from 2007Ks01 indicating $\Delta J=2$ or 0 seems inconsistent with $\Delta J=1$ from level scheme. $\Delta\text{IPDCO}=+0.04$ 4. DCO(Q)=0.49 6, $R_{\text{ADO}}=0.60$ 6 (2020Ra14).
1740	2.0 2	6087.6	13/2 ⁻	4347.8	9/2 ⁻	E2		I_γ : other: 2.3 2 at E=70 MeV. $\Delta\text{IPDCO}=+0.12$ 7.
1763	>9.7	1763.0	5/2 ⁺	0.0	3/2 ⁺	E2+M1		I_γ : other: >15.7 at E=70 MeV. $\Delta\text{IPDCO}=+0.13$ 5.
1786	1.8 4	7873.3	13/2 ⁺	6087.6	13/2 ⁻	D(+Q)	-0.6 6	I_γ : other: 0.1 1 at E=70 MeV. DCO(Q)=1.2 3 at E=88 MeV.
1862		10181.1	19/2 ⁻	8319.4	15/2 ⁻			
1946	10 1	7873.3	13/2 ⁺	5927.1	(11/2 ⁻)	E1+M2	+0.2 1	I_γ : other: 0.5 1 at E=70 MeV. DCO(Q)=1.3 2 at E=88 MeV. $\Delta\text{IPDCO}=+0.11$ 6. DCO(Q)=0.45 8, $R_{\text{ADO}}=0.59$ 8 (2020Ra14).
1985 @	2.0 4	5927.1	(11/2 ⁻)	3942.0	9/2 ⁺			I_γ : other: 1.6 6 at E=70 MeV.
2017 @&	7 1	10862.2?	19/2 ⁺	8845.1	17/2 ⁺			
2179	6.5 7	3942.0	9/2 ⁺	1763.0	5/2 ⁺	E2		I_γ : other: 11.1 6 at E=70 MeV. $\Delta\text{IPDCO}=+0.07$ 5. E_γ : from 2013Bi10 only.
2232		8319.4	15/2 ⁻	6087.6	13/2 ⁻			I_γ : other: 50 3 at E=70 MeV.
2244	61 2	5407.4	11/2 ⁻	3163.1	7/2 ⁻	E2		DCO(Q)=1.0 1 at E=70 MeV, 1.2 1 at E=88 MeV. $\Delta\text{IPDCO}=+0.04$ 2.
2312		6660.2	(11/2 ⁻)	4347.8	9/2 ⁻			
2391		12572.2	23/2 ⁻	10181.1	19/2 ⁻			
2400	<1	8487.8	15/2 ⁻	6087.6	13/2 ⁻	D(+Q)	+0.2 +3-2	E_γ : other: 2394 (2007Ks01). DCO(Q)=1.4 4 at E=88 MeV.
2466	22 2	7873.3	13/2 ⁺	5407.4	11/2 ⁻	E1+M2	-0.25 10	I_γ : other: 1.2 1 at E=70 MeV. DCO(Q)=2.4 3. $\Delta\text{IPDCO}=+0.21$ 6. DCO(Q)=0.51 8, $R_{\text{ADO}}=0.65$ 9 (2020Ra14).
2614		11459.2	21/2 ⁺	8845.1	17/2 ⁺			
2631 @&	<1	8558.2?	(13/2 ⁻)	5927.1	(11/2 ⁻)	D(+Q)	-0.2 +2-3	DCO(Q)=2.4 7 at E=88 MeV.
2646	37 2	2645.8	7/2 ⁺	0.0	3/2 ⁺	E2		I_γ : other: 47 2 at E=70 MeV. $\Delta\text{IPDCO}=+0.08$ 5. DCO(D)=1.7 2, $R_{\text{ADO}}=1.6$ 2 (2020Ra14).
2757		8845.1	17/2 ⁺	6087.6	13/2 ⁻			
2764		5927.1	(11/2 ⁻)	3163.1	7/2 ⁻			E_γ : from 2013Bi10 only.

Continued on next page (footnotes at end of table)

$^{12}\text{C}(^{28}\text{Si},\alpha p\gamma)$ **2007Ks01,2013Bi10,2020Ra14** (continued) $\gamma(^{35}\text{Cl})$ (continued)

E_γ [†]	I_γ [‡]	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. [#]	δ [#]	Comments
2912	<1	8319.4	15/2 ⁻	5407.4	11/2 ⁻			E_γ : other: 2917 (2007Ks01, tentative). Mult., δ : D(+Q) with $\delta=-0.1$ +3-2 if J(8319)=13/2 in 2007Ks01. DCO(Q)=1.9 6 at E=88 MeV.
3003 [@]	>1.1	3003.1	5/2 ⁺	0.0	3/2 ⁺			I_γ : other: >0.6 at E=70 MeV.
3080	<1	8487.8	15/2 ⁻	5407.4	11/2 ⁻			E_γ : other: 3074 (2007Ks01).
3163	100 5	3163.1	7/2 ⁻	0.0	3/2 ⁺	M2+E3	+0.16 1	I_γ : other: 100 2 at E=70 MeV. DCO(Q)=0.9 1 at E=70 MeV, 0.7 1 at 88 MeV. $\Delta\text{IPDCO}=-0.06$ 2.
3497		6660.2	(11/2 ⁻)	3163.1	7/2 ⁻			
3734		16306.4	(27/2 ⁻)	12572.2	23/2 ⁻			
4347 [@]	<1	4347.8	9/2 ⁻	0.0	3/2 ⁺			I_γ : other: <1 at E=70 MeV.

[†] From 2013Bi10, unless otherwise noted.

[‡] From 2007Ks01, measured at E(^{28}Si)=88 MeV at the NSC. Values are also available from E=70 MeV data of 2007Ks01 at TIFR facility, but less complete and are given under comments where available.

[#] Q and D+Q with $\delta(Q/D)$ are from $\gamma\gamma(\theta)(\text{DCO})$ in 2007Ks01 and magnetic/electric characters are from $\gamma\gamma(\text{lin pol})$ in 2007Ks01 where available and/or RUL where level T_{1/2} is measured, unless otherwise noted.

[@] From 2007Ks01 only.

[&] Placement of transition in the level scheme is uncertain.

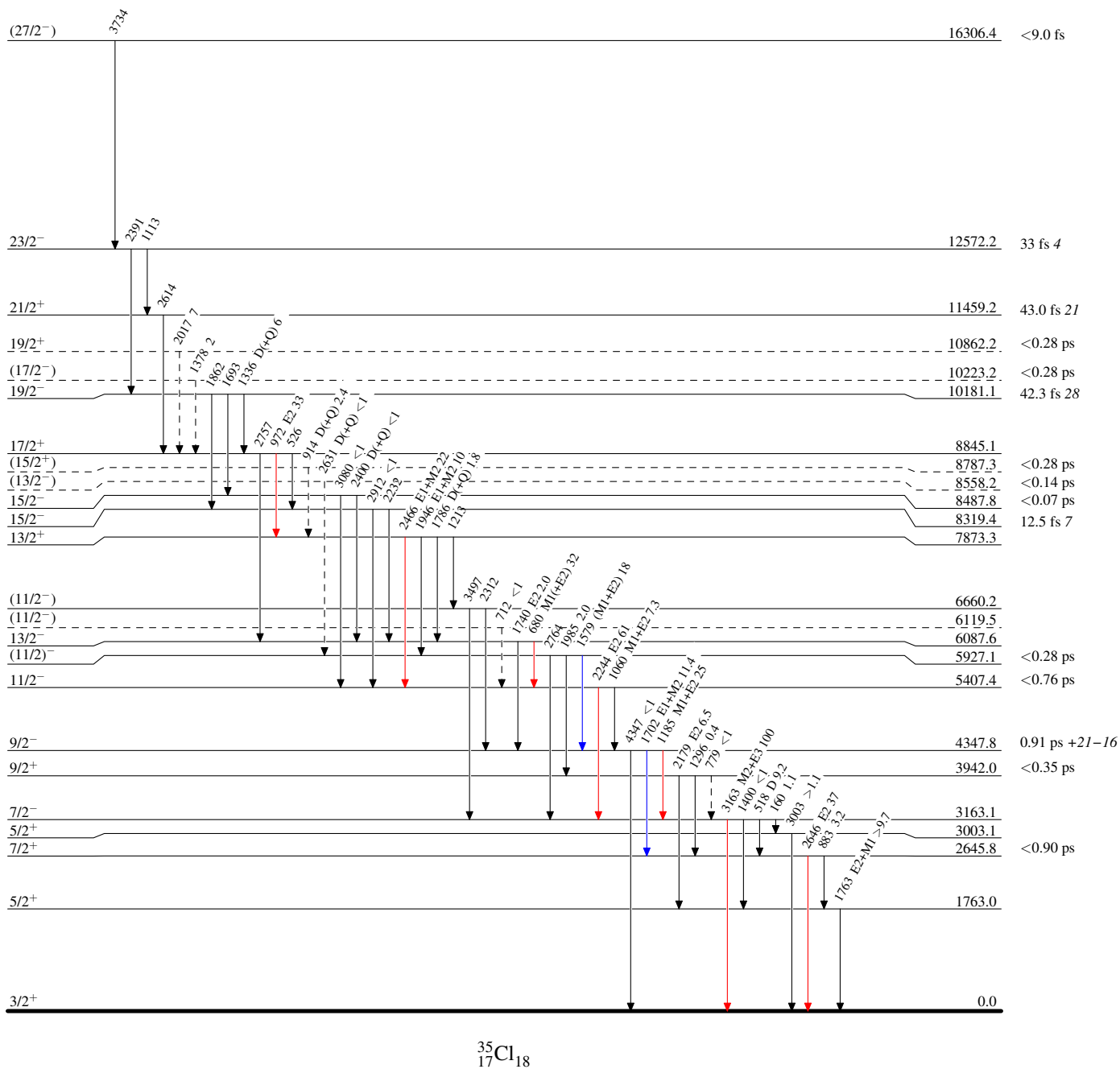
$^{12}\text{C}(^{28}\text{Si}, \alpha p \gamma)$ 2007Ks01, 2013Bi10, 2020Ra14

Legend

Level Scheme

Intensities: Relative I_γ

- $\longrightarrow$ $I_\gamma < 2\% \times I_\gamma^{\max}$
 $\longrightarrow$ $I_\gamma < 10\% \times I_\gamma^{\max}$
 $\longrightarrow$ $I_\gamma > 10\% \times I_\gamma^{\max}$
 $\longrightarrow$ γ Decay (Uncertain)



$^{12}\text{C}(^{28}\text{Si}, \alpha p \gamma)$ 2007Ks01, 2013Bi10, 2020Ra14

